

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

- Homework 4
  - Due **this Friday** 12/6 @5 pm
- Extra Credit Opportunities
  - Final Question Contribution
  - Homework 5
  - Final Exam Practice
- Course Evaluation
  - Please share your feedback and insights for me to improve my teaching methods and enhance the learning experience for future students. Thank you!

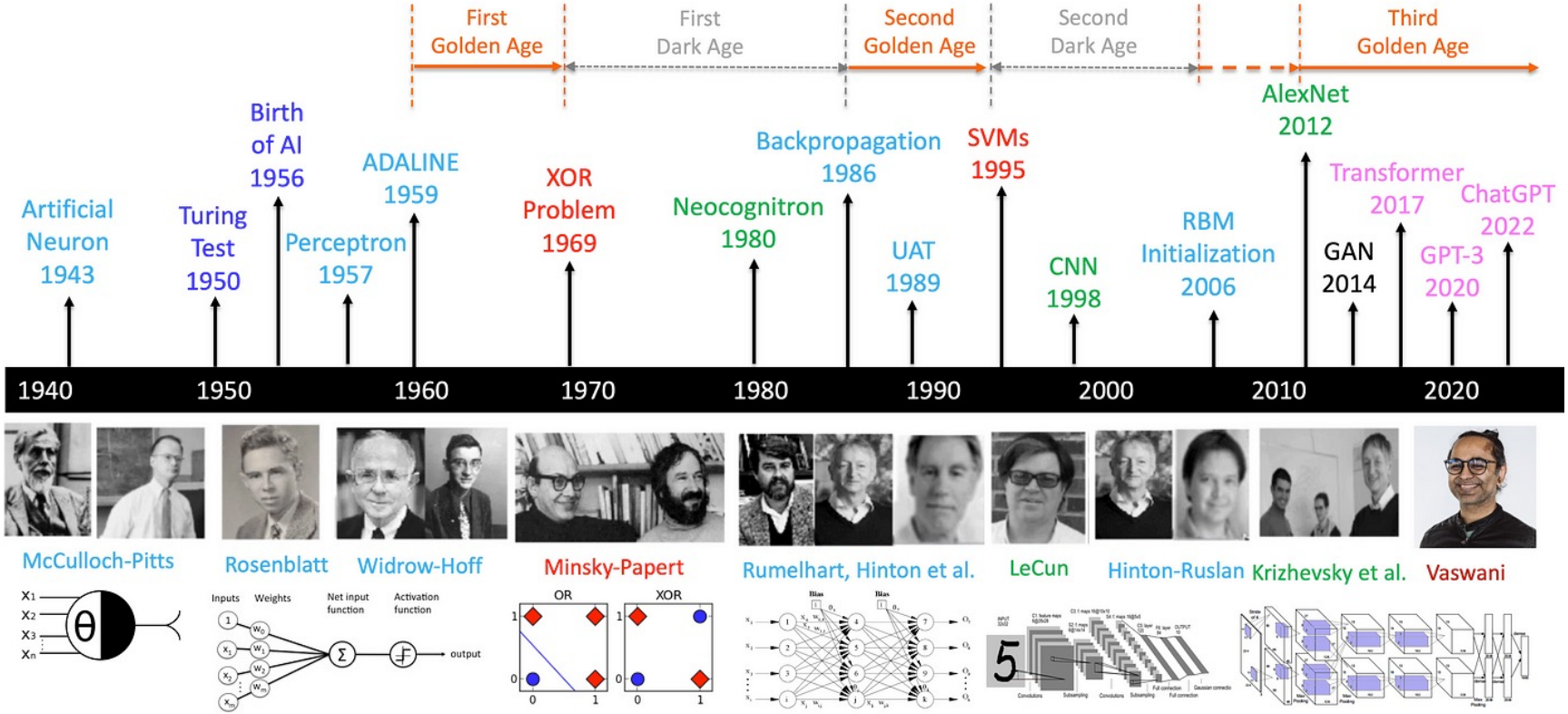
Test-Taking Strategies

- Show your work
  - Write out your thinking process (even on scratch paper)
  - Partial credit is often given for approach
  - Don’t skip multiple choice questions (if there is any)
- Cheat sheet
  - Organize information logically – don’t get overloaded
  - Prioritize content
  - Practice using your cheat sheet

This Lecture

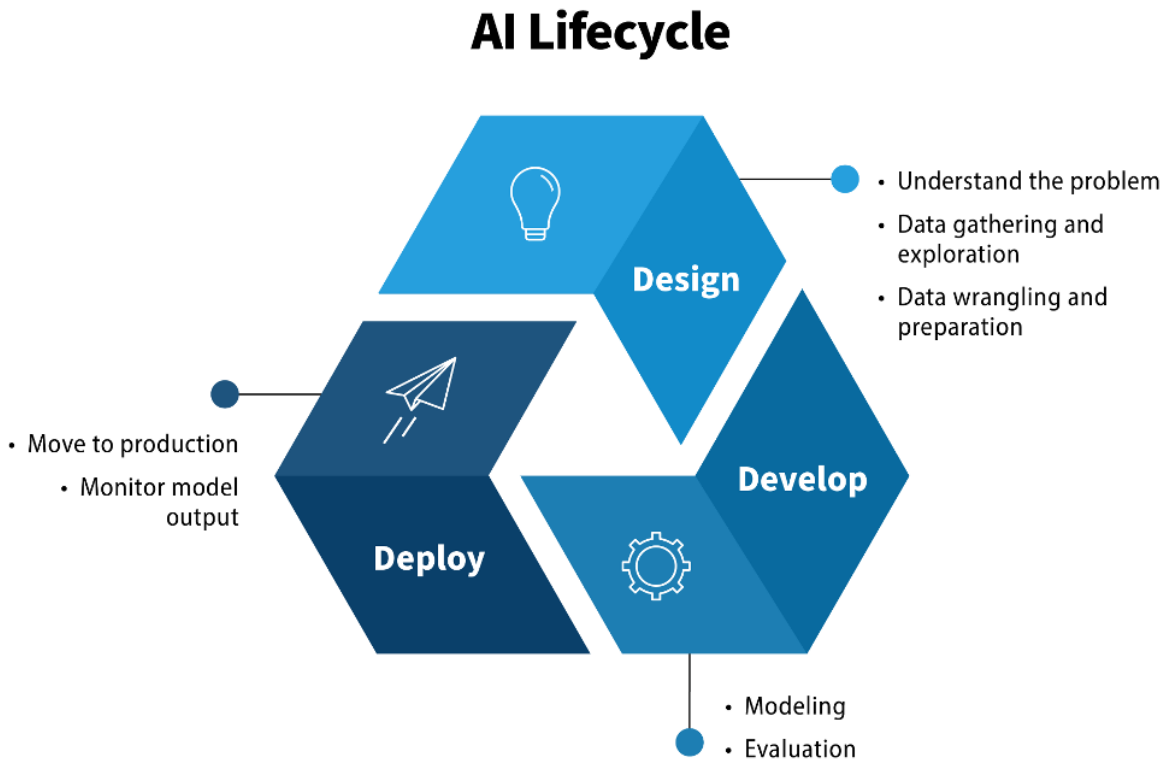
- History of AI
- AI Life Cycle
- Data Reliability
- Fairness
- Privacy
- Explainability
- Generative AI

A Brief History of AI with Deep Learning



Full Article: https://medium.com/@lmpo/a-brief-history-of-ai-with-deep-learning-26f7948bc87b

AI Life Cycle



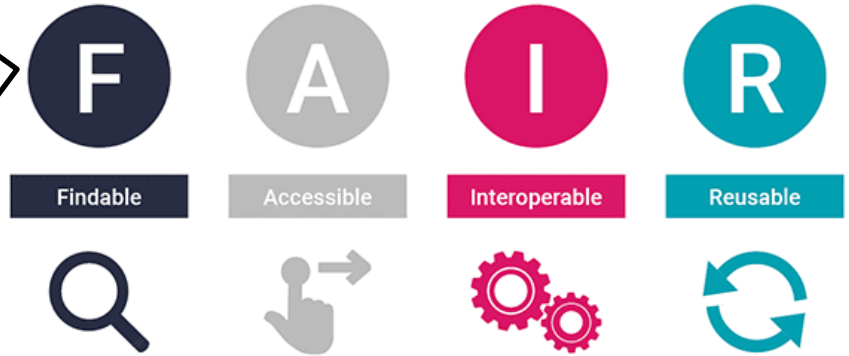
Good & Reliable Data

- What are the key characteristics of good & reliable data?



Data Collection and Quality Assessment

- Unique & persistent **identifiers** (e.g., DOI)
- Rich **metadata**
- **Indexed** in a searchable resource



- **Usage licenses**
- Meet domain-relevant **community standards**

- **Standardized protocols:** open, free, and universally implementable
- **Accessible Metadata**

- **Machine-readable**
- Use **formal**, **shared**, and broadly **applicable** languages

Fairness in AI

- How to define “fairness” in the context of AI systems?
- **Fairness:** absence of *prejudice or preference* for an **individual** or **group** based on their characteristics.

Individual Fairness



Similar individuals should receive similar outcomes.

vs.



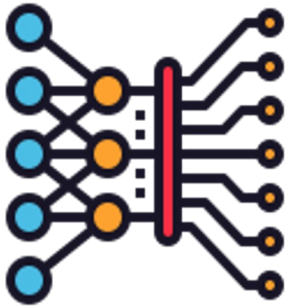
Group Fairness

Different groups (based on characteristics like race, gender) should receive similar outcomes.

Preventing Unfairness in AI



Data Collection & Exploration



Data Analysis & Modeling



Result Communication

Privacy in AI

- What's the most surprising or unsettling moment you've had when an app, website, or AI system seemed to know something about you that you didn't expect it to know?



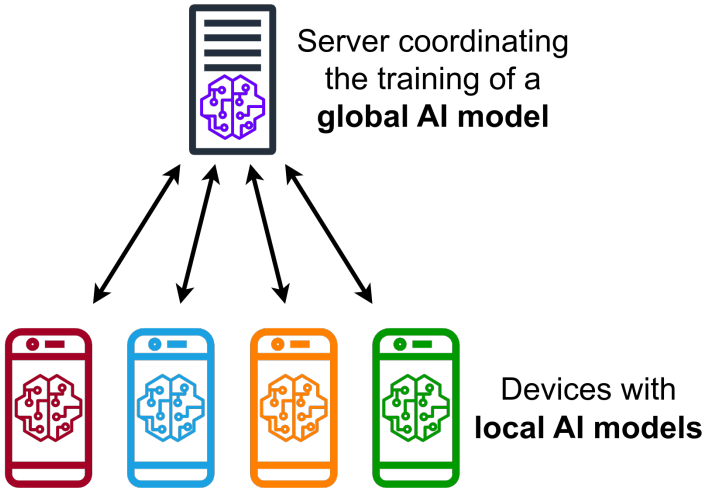
Data Privacy



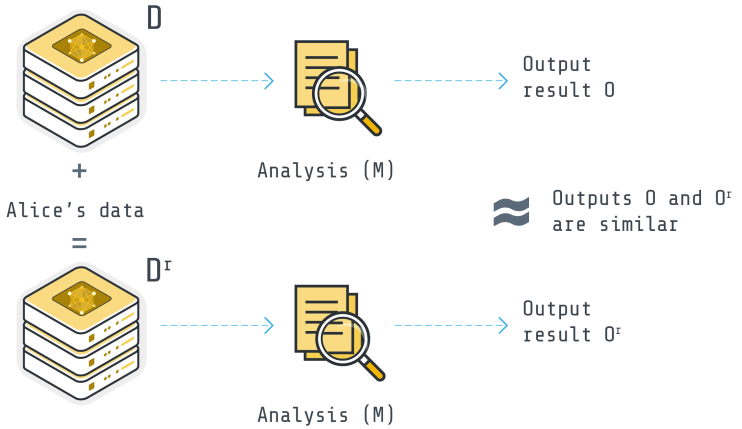
Video Link: https://www.youtube.com/watch?v=q55Tusc-rxc

Enhancing Privacy in AI

- Privacy-Preserving Learning Techniques



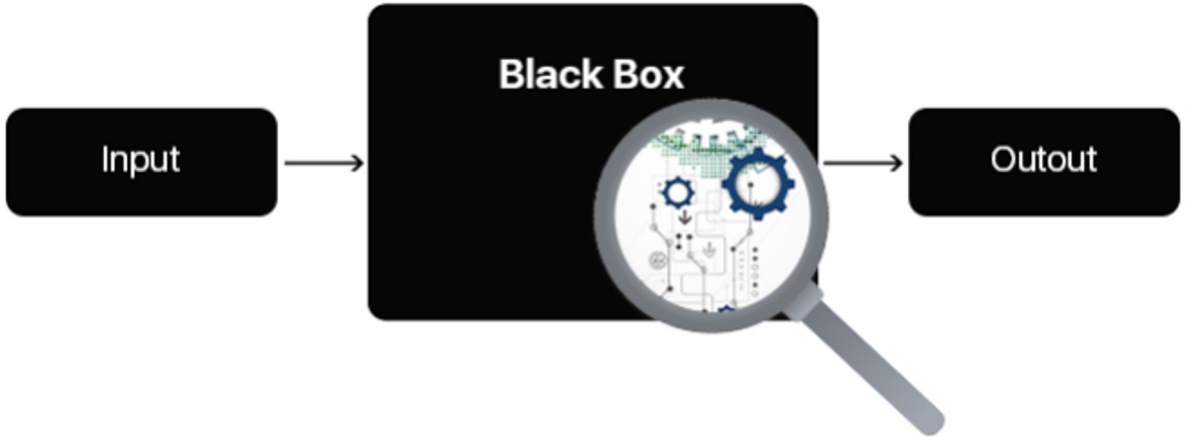
Federated Learning



Differential Privacy

Explainable AI

- Techniques and methods that make a model's **data**, **algorithms**, **decisions** understandable by people.

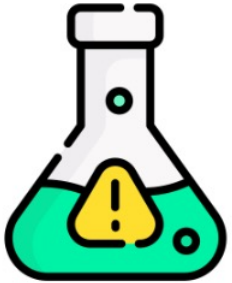


Explainable AI



Video Link: https://www.youtube.com/watch?v=It2Q1eK_Klc

Generative AI and its Ethical Issues



Toxicity

Harmful or discriminatory language or content



Hallucination

Factually incorrect content



Legal Aspects

Data Protection, Intellectual Property, and the EU AI Act

Environmental Concerns



Video Link: https://www.youtube.com/watch?v=MJQIQJYxey4

Next…

Q&A